

Vehicle Detection for Indian Urban Roads

YOLOv8n + UVH-26 Majority Voting

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30 / 30	30	8,000 / 2,000	58
Completed epochs	Best epoch	Train / validation	Tests passed

Proper baseline training and evaluation are complete.

Exit code 0; no early stopping. Fresh validation of best.pt and synchronized Apple MPS timing are recorded.
Phase 2 training has not started.

What this result means

A reproducible reference detector for 14 vehicle categories, including two-wheelers, three-wheelers and fine-grained car and commercial-vehicle types. The next research step is controlled improvement against this frozen reference.

Scope: subset validation, not full-dataset or independent test-set accuracy. Unselected-image acquisition and integrity remain incomplete. No production-readiness or real-time video claim.

Two different audit scopes

Evidence	Images	Boxes
Full MV annotation catalogue	26,646	316,220
Frozen local training subset	8,000	94,609
Frozen local validation subset	2,000	24,342

The catalogue audit checks JSON records and box metadata. The separate local audit decoded all 10,000 selected images, checked dimensions, paths, labels and SHA-256 hashes, and found no train-validation content overlap. All 14 classes are present. Full unselected-image pixel verification is unfinished.

Recovery preserved original files: a 1620-versus-1920 width mismatch in 803489.png could not be safely repaired. It and two visually degraded candidates were quarantined. Deterministic replacements were audited before freezing v2; no validation labels were changed after training.

The actual completed run

Setting	Executed value
Model / initialization	YOLOv8n / COCO pretrained
Epochs / best / stopping	30 / 30 / no early stop; exit 0
Image size / batch / seed	640 / 8 / 42
Optimizer	AdamW; lr0 0.000556; weight decay 0.0005
Warm-up / close mosaic	3 epochs / final 5 epochs
Device / effective AMP / workers	Apple M5 MPS / false / 0
Training wall duration	16029.859 s (4 h 27 min 10 s)
Runtime	Python 3.12.14; torch 2.14.0; Ultralytics 8.4.146

Both 6,228,714-byte checkpoints load with finite tensors and the correct taxonomy. Hashes and frozen configuration are recorded in the Phase 1 report. MPS nondeterminism warnings prevent a bitwise replay guarantee. Original run files remain unchanged.

Measured detection accuracy

Precision	Recall	F1*	Macro F1	mAP50	mAP50-95
0.6100	0.5435	0.5749	0.5394	0.5600	0.4584

2,000 validation images / 24,342 objects; best checkpoint from epoch 30. Fresh MPS evaluation: imgsz 640, batch 8, confidence floor 0.001, NMS IoU 0.7, max_det 300. AP spans IoU 0.50:0.05:0.95.

*F1 is 2PR/(P+R) using macro P/R. Macro F1 averages 14 class F1 scores. Common confidence 0.309309 maximizes smoothed mean class F1 at IoU 0.5. Neither is micro-F1.

Class	Precision	Recall	AP50	AP50-95
Hatchback	0.5621	0.6574	0.6345	0.5341
Sedan	0.5180	0.4936	0.5603	0.4884
SUV	0.4360	0.5691	0.4720	0.4115
MUV	0.4421	0.4291	0.4075	0.3645
Bus	0.7511	0.7124	0.7695	0.6430
Truck	0.6074	0.6506	0.6801	0.5454
Three-wheeler	0.8537	0.8391	0.9036	0.7403
Two-wheeler	0.8152	0.8016	0.8776	0.6393
LCV	0.6659	0.7080	0.7196	0.5790
Mini-bus	0.2179	0.1552	0.1891	0.1383
Tempo-traveller	0.6747	0.6062	0.6504	0.5694
Bicycle	0.5884	0.5502	0.5742	0.4220
Van	0.4074	0.4372	0.3626	0.3184
Others	1.0000	0.0000	0.0396	0.0242

Three-wheeler is strongest; Others and Mini-bus are weakest. Others has only 31 validation instances and zero recall at the F1 operating point; precision 1 is an empty-prediction interpolation convention. Mini-bus has only 58 instances. No test-set generalization or confidence intervals are established.

File-to-result timing

Stage	Mean ms	Median ms	p95 ms
end to end	31.357	31.732	34.359
inference	4.956	3.768	7.358
postprocess	2.466	1.404	4.532
preprocess	1.691	1.017	3.902

Inference-only: 201.77 FPS. End-to-end still images: 31.89 FPS. Rates are 1000 / mean milliseconds, not reciprocal median.

Apple M5, 24 GiB RAM, MPS float32, batch 1, imgsz 640 with rectangular letterboxing. Ten warm-ups excluded; 100 seed42 validation images timed. Each stage explicitly synchronizes MPS. Preprocessing converts BGR to RGB and normalizes; postprocessing uses class-aware NMS at conf 0.25 / IoU 0.7.

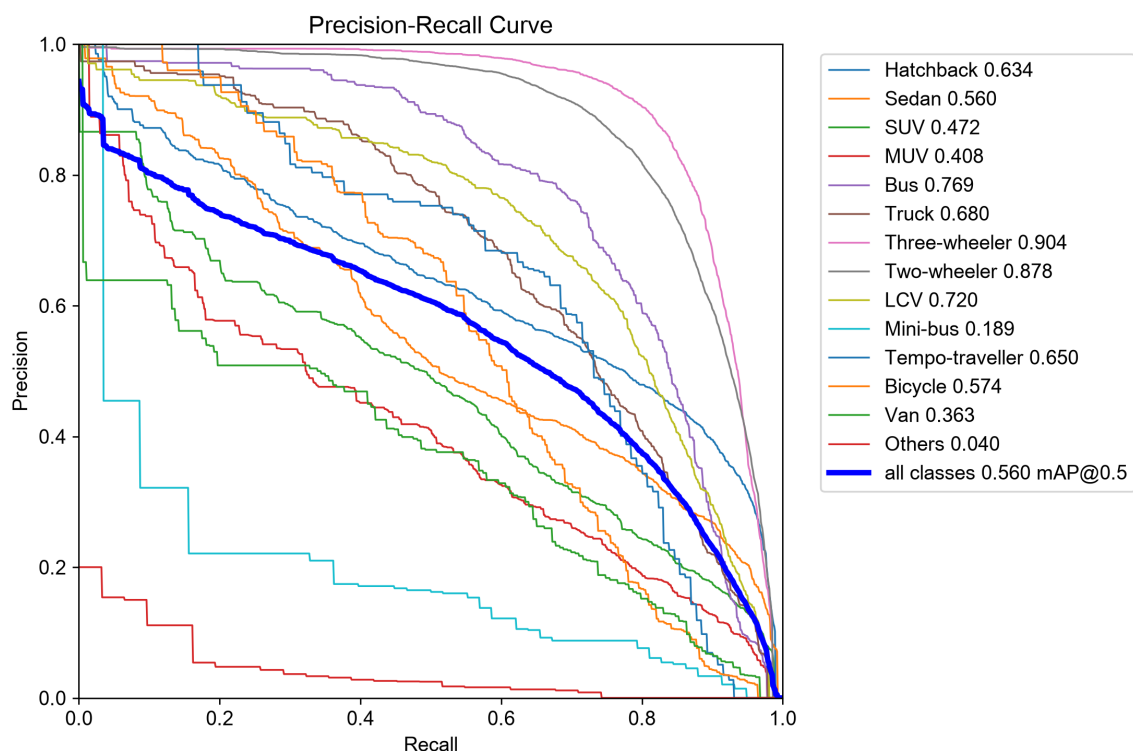
End-to-end includes local file read/decode, all processing, API overhead and synchronization barriers. It excludes model loading, drawing, video capture and UI. OS caching may affect file I/O. The library's unsynchronized MPS validation profiler is not used for these latency claims.

Ground-truth comparisons

Validation scene	Observed behavior
4232 / correct detections	10 correct-class matches, including truck and two-wheelers; two extra predictions remain.
10518 / small vehicle	Distant annotated two-wheeler missed; its box is about 65 square pixels after scaling to 640.
1364 / dense and occluded	48 correct-class matches out of 56 GT; 3 unmatched GT and many duplicate/extra low-confidence boxes. Pedestrian predicted as Bicycle.
4711 and 954 / rare classes	Mini-bus predicted Bus; construction vehicle (Others) predicted Truck. Unlabelled partial auto in 4711 is a separate source limitation.

101 diagnostic predictions generated; six GT/prediction pairs manually reviewed at conf 0.10. Matching is diagnostic IoU 0.5, not AP. Unmatched predictions can include unlabelled real vehicles. Occlusion-stratified or small-object AP has not been measured.

A reference for comparison



Standalone best-checkpoint precision-recall curves. Full per-class CSV, PR/F1/P/R plots and numeric confusion matrix accompany this PDF. The matrix uses confidence 0.001 and matching IoU 0.45, distinct from the selected F1 operating point.

Accuracy-speed trade-off

This compact detector is fast on still images, but rare-class recall, small vehicles and dense-scene duplicates limit usefulness. File/decode and API overhead dominate beyond inference. Larger models or inputs may improve accuracy but need controlled comparison; no such improvement is claimed yet.

Demonstrate the saved evidence

```
python3 scripts/show_progress.py
```

With dependencies installed: `python -m streamlit run app.py`

Run tests: `python -m pytest -q`

The portable ZIP contains source, configs, summarized results, tests, plots, this PDF and a presentation script. Dataset images, labels, checkpoints, full runs and local secrets are excluded. Read the saved subset Phase 2 gate and Git delivery status under reports/audit. Controlled Phase 2 work requires a Git checkpoint; no Phase 2 training was started.

Source: IISc AIM UVH-26, CC BY 4.0. Revision 59f82c57821e8a54dc40bc1f42e83909dbad0b70. Sharma et al., IISc Technical Report, November 2025, arXiv:2511.02563. All project metrics above come from included measured artifacts.